PROBLEM1:

PROBLEM STATEMENT:

Process marks for multiple subjects, check each subject's pass criteria, apply grace marks according to a specified policy, ensure marks do not exceed the maximum, and display the final result and classification.

ALGORITHM:

Step 1: Start.

Step 2: Read the number of subjects, maximum marks, pass marks, and grace-mark limit.

Step 3: Enter the marks obtained in each subject.

Step 4: Set `grace_used = 0` and initialize the total marks to zero.

Step 5: Check each subject's marks.

- If the marks are below the pass mark and the difference is within the grace-mark limit, apply grace marks if the policy allows.
- Ensure the final marks do not exceed the maximum marks.
- Check whether the subject is passed or failed.

Step 6: Calculate the total marks and percentage.

Step 7: If the student fails in any subject, declare the result as FAIL.

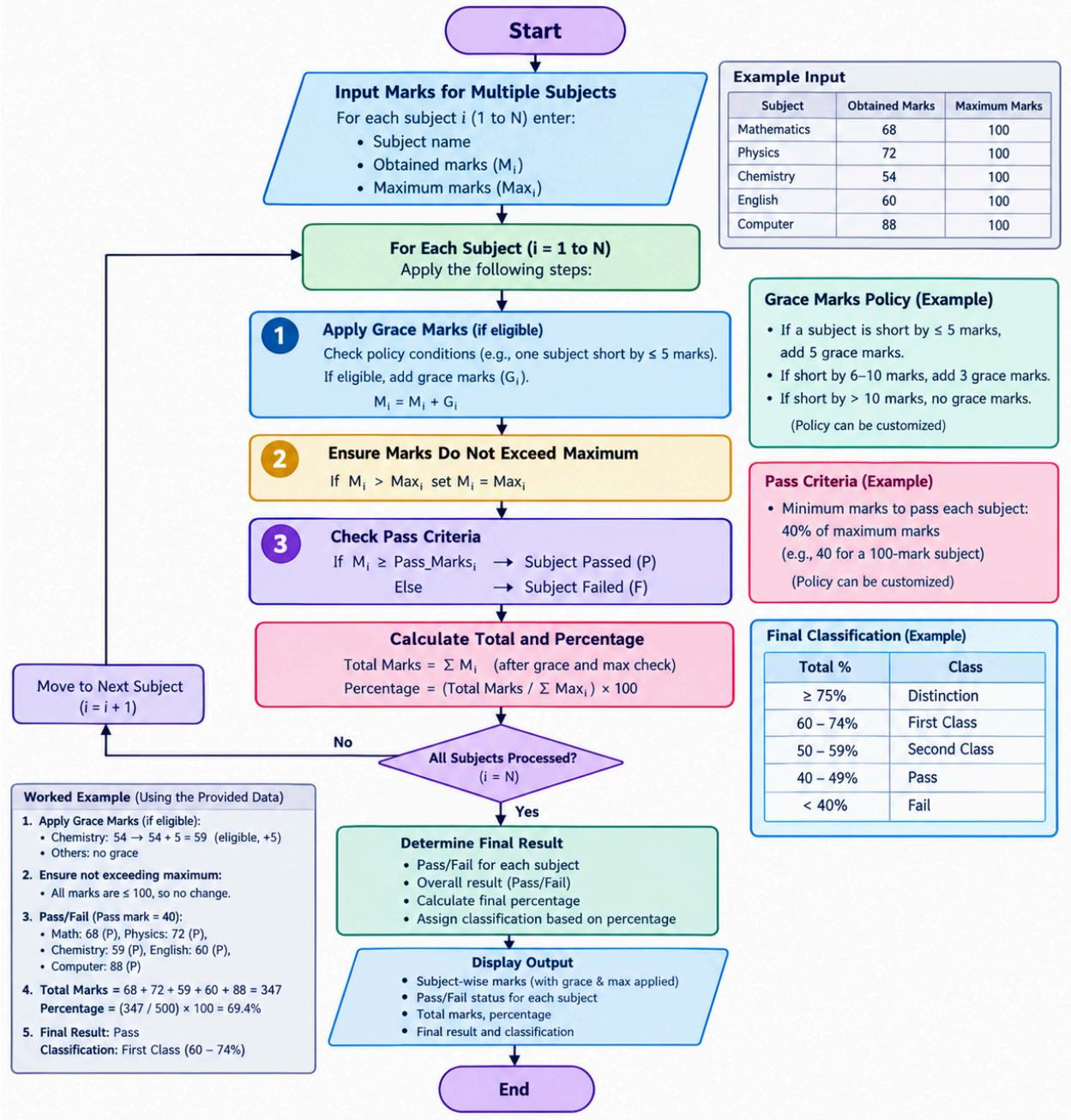
Step 8: Otherwise, determine the classification:

- Percentage ≥ 75%: Distinction
- Percentage ≥ 60%: First Class
- Percentage ≥ 50%: Second Class
- Otherwise: Pass

Step 9: Display the final marks, total marks, percentage, result, and classification.

Step 10: Stop.

FLOWCHART:



PSEUDO CODE:

BEGIN

INPUT number_of_subjects
INPUT maximum_marks, pass_mark, grace_limit

```
total ← 0
grace_used ← FALSE
failed ← FALSE

FOR each subject from 1 to number_of_subjects DO

   INPUT marks

   IF marks < pass_mark AND
     (pass_mark - marks) <= grace_limit AND
     grace_used = FALSE THEN

      marks ← MIN(marks + grace_limit, maximum_marks)
      grace_used ← TRUE

   END IF

   IF marks < pass_mark THEN
      failed ← TRUE
   END IF

   total ← total + marks
   DISPLAY marks

END FOR

percentage ← (total / (number_of_subjects * maximum_marks)) * 100

DISPLAY total
DISPLAY percentage

IF failed = TRUE THEN
   DISPLAY "FAIL"
   DISPLAY "Classification: Not Applicable"
ELSE
   IF percentage >= 75 THEN
      DISPLAY "Distinction"
   ELSE IF percentage >= 60 THEN
      DISPLAY "First Class"
   ELSE IF percentage >= 50 THEN
      DISPLAY "Second Class"
   ELSE
      DISPLAY "Pass"
   END IF
```

END IF

END

PROGRAM:

```
marks = [32, 67, 80, 34, 90]
maximum_marks = 100
pass_mark = 35
grace_limit = 5

final_marks = []
grace_used = 0

for mark in marks:
    if mark < pass_mark and pass_mark - mark <= grace_limit and grace_used == 0:
        mark = min(mark + grace_limit, maximum_marks)
        grace_used = 1

    final_marks.append(mark)

print("Final Marks:", final_marks)

if all(mark >= pass_mark for mark in final_marks):
    total = sum(final_marks)
    percentage = total / len(final_marks)

    if percentage >= 75:
        result = "Distinction"
    elif percentage >= 60:
        result = "First Class"
    elif percentage >= 50:
        result = "Second Class"
    else:
        result = "Pass"

    print("Total Marks:", total)
    print("Percentage:", round(percentage, 2))
    print("Classification:", result)
else:
    print("Result: FAIL")
    print("Classification: Not Applicable")
```

OUTPUT:

Final Marks: [37, 67, 80, 34, 90]
Result: FAIL
Classification: Not Applicable